Unit 1

Assignment

Long Vy

Purdue Global

IN450 – Advanced Software Development Using Python

Brian Myers

March 28, 2026

MEMORANDUM

TO: Professor Brian Myers

FROM: Long Vy, Development Team Lead

DATE: March 28, 2026

RE: Proposal for Adoption of Software Configuration Management (SCM) System

Introduction

The purpose of this memo is to present a formal recommendation for the use of a Software Configuration Management (SCM) system for our team. Our team currently consists of three software developers and two quality testers, with additional read-only access required for project management oversight. Software configuration management is a management and tracking system for software products that keeps development consistent and ensures integrity while coordinating the work of multiple team members and allowing them to track changes, remain in sync, and stay on task (Coursera Staff, 2025). As our workload grows and the number of active projects increases, establishing a formal SCM system has become a critical operational requirement.

This proposal outlines the business case for SCM adoption comparing two leading platforms, GitHub and Microsoft Azure DevOps, and concludes with a justified recommendation suited to our team size, workflow, and budget.

Why do we Need Software Configuration Management

When software is developed collaboratively, the product undergoes continuous changes during the maintenance phase, and many team members work together writing programs toward common goals, generating a growing collection of configuration items that must be handled effectively (Savajia, 2021). SCM handles issues with reproducing past builds that result from ineffective tracking, configuration discrepancies, and code overwrites (Coursera Staff, 2025). Without a formal SCM system in place, teams risk losing version history, overwriting each other's contributions, and introducing untracked errors that grow increasingly costly to resolve over time.

Managing Change Across Different Projects

SCM is a discipline that identifies change, monitors and controls change, ensures the proper implementation of changes made to items, and audits and reports on those changes (Savajia, 2021). Over the course of a year, our team may be engaged in multiple concurrent projects, many of which have not yet been scoped or identified. Both GitHub and Azure DevOps support the creation of multiple project repositories under a single organization account, allowing the team to provision isolated environments for new projects quickly without disrupting work already in progress.

Collaboration Between Dev Team and Quality Assurance

SCM is used not only by software developers but also by quality assurance testers, who rely on configuration management-assured, up-to-date versions of software when testing for bugs (Coursera Staff, 2025). GitHub supports team workflows with code reviews, discussions, and pull requests, providing a structured process for proposing, reviewing, and merging changes into the codebase (GeeksforGeeks, 2026). For our two QA testers, this ensures they are always pulling stable, reviewed builds for testing while developers continue work in separate branches, keeping the testing environment clean and reducing the risk of introducing untested code into production.

Project Manager Visibility

Project managers play a direct role in SCM by ensuring team members follow guidelines and updating stakeholders on progress (Coursera Staff, 2025). Azure DevOps provides customizable dashboards displaying real-time project data including build status, test results, and work item queries, giving stakeholders a central hub to monitor team progress without requiring developer-level access (Chcomley, n.d.). GitHub similarly offers project boards and repository insights that support stakeholder visibility into sprint status and milestone completion.

Versioning and Branching for Custom Client Features

A key scenario our team will face is delivering custom or paid features requested through the Marketing Department. Version control systems allow teams to track and manage the software developed based on a baseline approved version, noting any deviations and modifications as they occur (Coursera Staff, 2025). GitHub's branching capability allows teams to develop features or fixes in parallel without impacting the main branch, and pull requests enable a structured review and merge process before changes reach production (GeeksforGeeks, 2026). A dedicated branch can therefore be maintained for any client-specific feature, keeping it isolated until it has been reviewed, approved, and is ready to release. Azure DevOps offers similar branch policies with enforced approval requirements (Chcomley, n.d.), which means both platforms give our team the controls needed to deliver custom work without compromising the stability of the main codebase.

System Comparison: GitHub vs. Microsoft Azure DevOps

Two industry-leading SCM platforms have been evaluated for this proposal are GitHub and Microsoft Azure DevOps. Both are cloud-based, Git-compatible platforms with mature toolsets, but they differ in workflow, ecosystem integration, and cost structure.

GitHub

GitHub is a web-based platform that hosts Git repositories and provides tools for version control, collaboration, and project management, designed to help teams work efficiently on projects of any size (GeeksforGeeks, 2026). Its core features include repositories for storing project files and complete version history, branching for parallel development, pull requests for proposing and reviewing changes, issues for tracking bugs and enhancements, and GitHub Actions for automating CI/CD workflows (GeeksforGeeks, 2026). New repositories can be created quickly through the web interface, and each includes its own issue tracker, wiki, branch management tools, and configurable access controls, making it straightforward to onboard new projects at any point in the year.

GitHub pricing is straightforward for small teams. The GitHub Free plan supports unlimited public and private repositories with core collaboration features. The GitHub Team plan, at $4 per user per month, adds required reviewers, code owners, protected branches, and 3,000 CI/CD minutes per month. For our team of five plus one project manager, the Team plan would cost approximately $24 per month or $288 per year.

Microsoft Azure DevOps

Azure DevOps is a cloud-based platform that provides integrated tools for software development teams, encompassing everything needed to plan work, collaborate on code, build applications, test functionality, and deploy to production (Chcomley, n.d.). Its core services include Azure Boards for Agile work item tracking and sprint planning, Azure Repos for Git-based source control, Azure Pipelines for CI/CD automation, Azure Test Plans for manual and automated testing, and Azure Artifacts for package management (Chcomley, n.d.). These services are designed to integrate across the full development lifecycle, from initial planning through final deployment.

Azure DevOps supports both Git and Team Foundation Version Control (TFVC), offering flexibility for teams with existing workflows. Branch policies can be enforced organization-wide, requiring pull request approvals and passing build checks before any merge to a protected branch, directly supporting the team's need to manage custom client feature branches with appropriate controls (Chcomley, n.d.). Customizable dashboards and real-time reporting also give project managers clear visibility into team progress (Coursera Staff, 2025), which means stakeholders can stay informed on sprint status and milestones without needing direct access to the codebase.

On pricing, Azure DevOps offers a free Basic plan for up to five users, which includes Azure Boards, Azure Repos, and Azure Pipelines with 1,800 CI/CD minutes per month. A sixth user, our project manager, would require a Basic license at $6 per user per month, bringing the total estimated cost to $6 per month or $72 per year. This makes Azure DevOps the more cost-effective option at our current team size.

Cloud-Based Solutions: Advantages and Disadvantages

Both platforms are fully cloud-hosted, which introduces important considerations for our team. The advantages of cloud-based SCM are significant. Choosing the right SCM tools requires evaluating the reputation of the vendor, the cost of the software, licensing costs, and the team's budget for training (Coursera Staff, 2025), and cloud-hosted platforms address most of these concerns by eliminating the need for on-premises infrastructure entirely. Both GitHub and Azure DevOps are accessible from any location or device, require no on-premises servers to maintain, and include automatic backups and redundancy. Azure DevOps is built on Azure's global infrastructure with a 99.9% service level agreement and enterprise-grade security including Microsoft Entra ID integration and compliance certifications (Chcomley, n.d.), making it a reliable choice for teams that need consistent uptime and data protection.

The primary disadvantages of cloud-based SCM include dependency on internet connectivity for core operations, potential data sovereignty concerns for organizations subject to strict compliance requirements, and the low but non-zero risk of service disruptions. Additionally, SCM tools must be evaluated to ensure they offer the features and integrations needed to support existing workflows, and that teams are provided adequate training to use them effectively (Coursera Staff, 2025). These considerations should be addressed before full adoption through a structured onboarding and training plan.

Recommendation and Justification

Based on the analysis above, this proposal recommends GitHub Team as the SCM platform for our development team.

GitHub's combination of an intuitive user friendly interface, powerful branching model, and structured pull request workflow makes it the strongest fit for our team's development and QA processes. Its branching and version control capabilities directly address the custom client feature scenario, where developers can maintain a dedicated branch for any paid client request, isolated from the main product line until it has been reviewed, approved, and is ready to merge (GeeksforGeeks, 2026). The platform also maps cleanly to the five core SCM components, identification, version control, change management, configuration status accounting, and auditing (Coursera Staff, 2025), which means adopting GitHub gives our team a structure that aligns with industry-recognized best practices from day one.

While Azure DevOps offers a lower entry cost and strong integrated tooling, its full value is best realized in organizations already embedded in the Microsoft ecosystem. For a lean development team prioritizing developer experience and broad community support, GitHub provides a more accessible workflow and a larger ecosystem of integrations. At $24 per month for our full team of six, the cost is well within reason for the productivity and risk-management value the platform delivers.

It is recommended that the team adopt GitHub Team, create a shared organization account, and establish a standardized branching convention within the first sprint of adoption.

Conclusion

Software configuration management is the management and tracking system that keeps development consistent and ensures integrity across the entire software development lifecycle (Coursera Staff, 2025). This memo has outlined the business case for SCM adoption, compared GitHub and Azure DevOps across functionality, branching and versioning, project scalability, cloud considerations, and pricing, and recommended GitHub Team as the platform that best meets our team's current and anticipated needs. Implementing a formal SCM system will equip our developers, QA testers, and project manager with the tools needed to work efficiently, maintain accountability, and deliver quality software.

Approval to proceed with account setup and repository initialization is respectfully requested at the earliest opportunity. Please do not hesitate to reach out with any questions or requests for additional information.

Appendix: GitHub Account Screenshots

The following screenshots confirm successful account creation and sign-in to GitHub using a Purdue Global university email address (longvy@student.purdueglobal.edu).

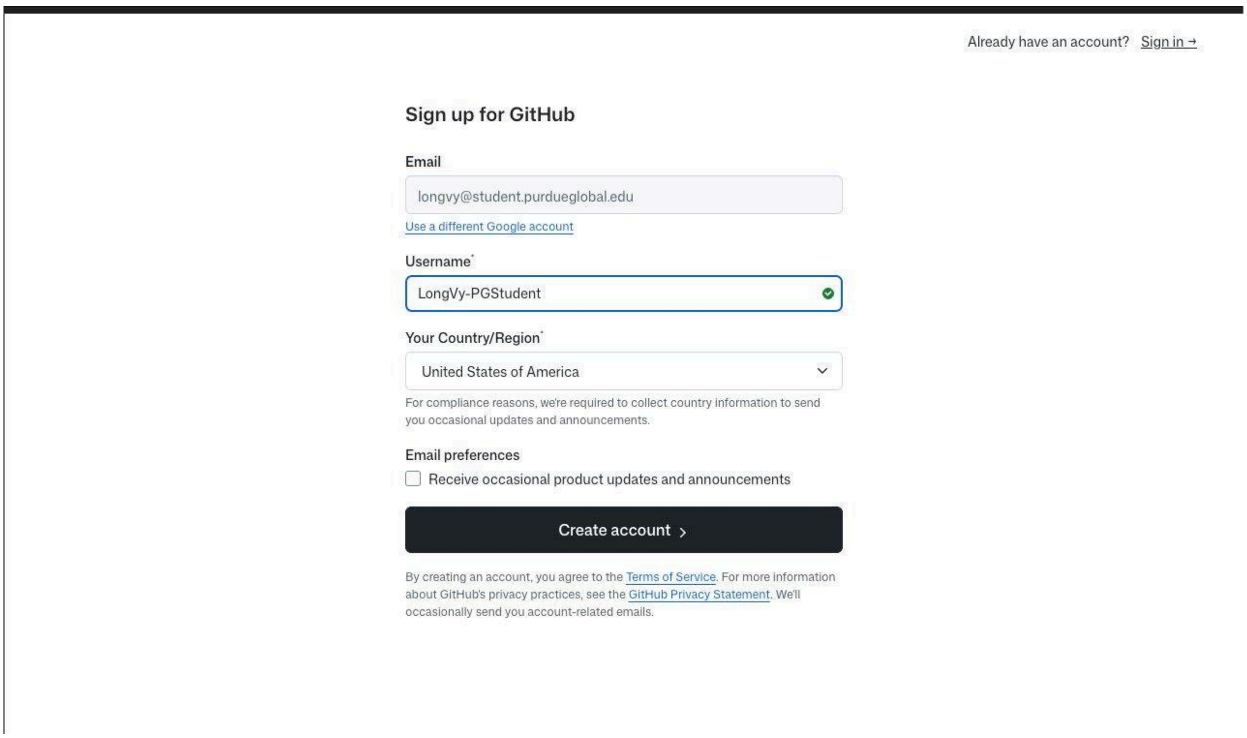

Figure 1. GitHub account registration using university email longvy@student.purdueglobal.edu

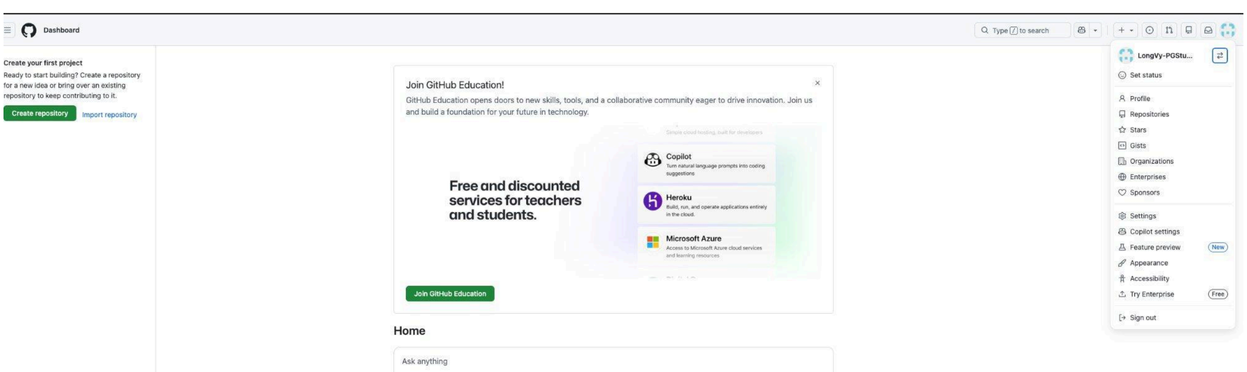

Figure 2. GitHub dashboard confirming successful sign-in as LongVy-PGStudent

References

Chcomley. (n.d.). *What is azure devops? - azure DevOps*. Azure DevOps | Microsoft Learn. https://learn.microsoft.com/en-us/azure/devops/user-guide/what-is-azure-devops?view=azure-devops

Coursera Staff. (2025, September 30). *Software configuration management: Streamlining your development workflow*. Coursera. https://www.coursera.org/articles/software-configuration-management

GeeksforGeeks. (2026, February 26). *Introduction to github and its workflow*. https://www.geeksforgeeks.org/git/what-is-github-and-how-to-use-it/

Savajia1, V. (2021, February 21). *What is software configuration management ? how we can leverage ServiceNow for SCM.* What is Software Configuration Management ? How we... -ServiceNowCommunity. https://www.servicenow.com/community/developer-articles/what-is-software-configuration-management-how-we-can-leverage/ta-p/2307390